

The 3-majority dynamics on the complete graph: a formalization-oriented proof of the $O(\log n)$ consensus bound

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Abstract

We give a complete, elementary proof that the *3-majority* opinion-dynamics process on n fully-mixing agents, each holding one of two opinions, reaches consensus within $O(\log n)$ rounds with probability $1 - O(1/n)$, provided the initial imbalance between the two opinions exceeds a fixed constant fraction of n . The proof is organized for formalization in Lean 4 + Mathlib, mirroring the companion *rumor-spreading* development (`rumor_spread/`): the probability space is finite and uniform, and the only analytic ingredient beyond finite sums is a single self-contained exponential-moment (Chernoff-type) concentration lemma for sums of independent $\{0, 1\}$ -valued random variables, proved from the elementary bound $1 + x \leq e^x$ — no measure theory, no PMF/kernels, no appeal to Mathlib’s probability library. This is a genuinely different regime from the rumor-spreading protocol: the underlying opinion count is *not* monotone (it can go up or down each round), so — unlike the push protocol — a counting argument over “good rounds” does not suffice, and honest concentration around the process’s mean trajectory is unavoidable.

This argument has been *fully formalized* (no `sorry`) in the accompanying Lean development (`ThreeMajority/`); the main theorem is `ThreeMajority.majority3_consensus_whp`.

1 The model

Fix $n \geq 2$ and let $V = \{0, 1, \dots, n-1\}$ be the set of agents, each holding one of two opinions, which we call 1 (“red”) and 0 (“blue”). Write $I \subseteq V$ for the set of agents currently holding opinion 1; the complement $V \setminus I$ holds opinion 0.

Definition 1.1 (Round randomness). *A round configuration is a function $\omega: V \rightarrow V \times V \times V$ assigning to every agent v an ordered triple of samples $\omega(v) = (\omega_1(v), \omega_2(v), \omega_3(v))$, each drawn independently and uniformly from V (sampling with replacement: an agent may sample itself, and may sample the same agent more than once — this is harmless since a majority of 3 Boolean values is always well defined, with no ties). Let $R_n = V^{3V}$ denote the (finite) set of round configurations, with the uniform probability measure.*

Definition 1.2 (Majority step). *For an opinion-1 set $I \subseteq V$ and $\omega \in R_n$, define*

$$\text{step}(I, \omega) = \{v \in V : \text{at least two of } \omega_1(v), \omega_2(v), \omega_3(v) \text{ lie in } I\}.$$

That is, each agent independently samples 3 agents and adopts the majority of their current opinions.

Definition 1.3 (Process). *Fix a horizon $T \in \mathbb{N}$ and an initial set $I_0 \subseteq V$. The sample space is $\Omega = R_n^T$ with the uniform (i.i.d.) measure. For $\omega = (\omega_0, \dots, \omega_{T-1}) \in \Omega$ define $I_{t+1}(\omega) = \text{step}(I_t(\omega), \omega_t)$. We write $X_t = |I_t|$ and $U_t = n - X_t$.*

Lemma 1.4 (Monotone coupling). *For every round configuration $\omega \in R_n$, the map $I \mapsto \text{step}(I, \omega)$ is monotone: $I \subseteq I'$ implies $\text{step}(I, \omega) \subseteq \text{step}(I', \omega)$.*

Proof. If $v \in \text{step}(I, \omega)$ then at least two of v 's three samples lie in $I \subseteq I'$, hence lie in I' , so $v \in \text{step}(I', \omega)$. $\square$

Lemma 1.4 is purely combinatorial (no probability involved), and is the single structural fact that lets us reduce “what happens starting from an arbitrary set of size $\geq M$ (resp. $\leq M$)” to “what happens starting from a *specific* reference set of size exactly M ,” using the *same* round configuration ω on both. Concretely: if $|I| \geq M$, choose any $I' \subseteq I$ with $|I'| = M$; then $|\text{step}(I, \omega)| \geq |\text{step}(I', \omega)|$ for every ω , so any *lower*-tail bound proved for the reference set of size M transfers to I . Symmetrically, if $|I| \leq M$, embed I into a reference $I' \supseteq I$ of size M ; then $|\text{step}(I, \omega)| \leq |\text{step}(I', \omega)|$ for every ω , so *upper*-tail bounds transfer. We call this the *reference-embedding* trick; it replaces the optional-stopping/martingale machinery that would otherwise be needed to control a non-monotone process that must stay inside a region for many rounds.

2 Main theorem

Throughout, $\log$ denotes the natural logarithm.

Theorem 2.1 (3-majority reaches consensus in $O(\log n)$ rounds w.h.p.). *There is an explicit constant N_0 such that for all $n \geq N_0$ and all $I_0 \subseteq V$ with $10|I_0| \geq 6n$ (i.e. the initial fraction of opinion-1 agents is at least $3/5$), setting*

$$T = 10 + \lceil 6 \log n \rceil + 2,$$

we have

$$\mathbb{P}[I_T = V] \geq 1 - \frac{C}{n}$$

for an explicit absolute constant C . In particular consensus is reached within $O(\log n)$ rounds with high probability.

Remark 2.2. *The hypothesis $|I_0| \geq 0.6n$ is a genuine simplification relative to the sharp threshold in the literature. It is known (Doerr–Goldberg–Minder–Sauerwald–Scheideler 2011; Becchetti–Clementi–Natale–Pasquale–Silvestri, SODA 2015) that an initial imbalance of only $\Theta(\sqrt{n \log n})$ already suffices for $O(\log n)$ -round consensus w.h.p. (and this is essentially tight: an imbalance of $o(\sqrt{n})$, as arises e.g. from an unbiased coin flip per agent, is not enough — the process can get stuck near the unstable balanced point $|I| = n/2$). Reaching that sharp threshold requires tracking the process on the scale of its own standard deviation throughout the entire trajectory (a genuinely delicate multi-scale concentration argument). We instead assume a constant-fraction initial imbalance, which keeps every step of the proof below to a single, reusable, elementary concentration lemma (Section 4) applied at a handful of explicit parameter values. The qualitative content — $O(\log n)$ rounds, probability $1 - O(1/n)$ — is unaffected; only the very final “distance from the constant C to the actual constant depends on the assumed initial threshold, and is not tight” should be taken as an implementation choice, not a mathematical necessity.*

The proof has two phases, exactly as in the push protocol, but for a structurally different reason: *growth* (Section 5), where the opinion-1 majority self-amplifies from 60% to 75% of the population; and *saturation* (Section 6), where the last dissenting agents are eliminated. Because the underlying process is not monotone (an unlucky round can shrink I_t), *both* phases require the same concentration tool — unlike the push protocol, where only the saturation phase needed a (much weaker) expectation argument and the growth phase could get by on a soft “good rounds accumulate” counting trick. This is the key mathematical difference from the rumor-spreading proof.

3 The majority map

Write $x = X_t/n \in [0, 1]$ for the fraction of opinion-1 agents.

Lemma 3.1 (Exact one-round drift). *Conditionally on $X_t = m$ (i.e. on a specific set I with $|I| = m$), the next count $X_{t+1} = |\text{step}(I, \omega)|$ is exactly a sum of n independent (but not identically distributed, in general) $\{0, 1\}$ -valued random variables — one per agent, since each agent's three samples are independent of every other agent's — with*

$$\mathbb{E}[X_{t+1}] = np(x), \quad p(x) := 3x^2 - 2x^3, \quad x = m/n.$$

Proof. Fix agent v . Its three samples are i.i.d. uniform on V , so each lands in I independently with probability x ; the majority of three i.i.d. Bernoulli(x) trials is 1 with probability $\binom{3}{2}x^2(1-x) + \binom{3}{3}x^3 = 3x^2(1-x) + x^3 = 3x^2 - 2x^3$. Sum over $v \in V$ (linearity of expectation) to get $\mathbb{E}[X_{t+1}] = np(x)$; independence across agents is immediate since each agent's samples are drawn from a disjoint block of the product space $R_n = V^{3V}$. $\square$

Lemma 3.2 (Algebraic identities for p). *For all $x \in [0, 1]$:*

$$p(x) - x = x(1-x)(2x-1), \tag{1}$$

$$p(1-x) = 1-p(x), \tag{2}$$

$$1-p(x) = (1-x)^2(1+2x). \tag{3}$$

Consequently, writing $\beta = x - \frac{1}{2}$ for the signed bias:

$$p\left(\frac{1}{2} + \beta\right) - \frac{1}{2} = \beta\left(\frac{3}{2} - 2\beta^2\right). \tag{4}$$

Proof. All three are polynomial identities, checked by expanding $p(x) = 3x^2 - 2x^3$; (4) is (1) rewritten at $x = \frac{1}{2} + \beta$: $p(x) - x = x(1-x)(2x-1)$, and at $x = \frac{1}{2} + \beta$, $x(1-x) = \frac{1}{4} - \beta^2$ and $2x-1 = 2\beta$, so $p(x) - x = 2\beta(\frac{1}{4} - \beta^2) = \frac{1}{2}\beta - 2\beta^3$; adding $x - \frac{1}{2} = \beta$ gives $p(x) - \frac{1}{2} = \frac{3}{2}\beta - 2\beta^3 = \beta(\frac{3}{2} - 2\beta^2)$. $\square$

Corollary 3.3 (Growth region). *For $0 \leq \beta \leq \frac{1}{4}$: $p(\frac{1}{2} + \beta) - \frac{1}{2} \geq \frac{5}{4}\beta$.*

Proof. By (4), the factor $\frac{3}{2} - 2\beta^2 \geq \frac{3}{2} - 2 \cdot \frac{1}{16} = \frac{11}{8} > \frac{5}{4}$ for $\beta \leq \frac{1}{4}$. $\square$

Corollary 3.4 (Saturation region). *For $0 \leq u \leq \frac{1}{4}$: $n(1-p(1-u)) \leq \frac{5}{8}un$, i.e. writing $U = un$ for an uninformed-count on scale $u \leq n/4$: the drift map $g(U) := n(1-p(1-U/n)) = \frac{U^2}{n}\left(3 - \frac{2U}{n}\right)$ satisfies $g(U) \leq \frac{5}{8}U$.*

Proof. By (3) with $x = 1-u$: $1-p(1-u) = u^2(3-2u)$. The map $u \mapsto u(3-2u)$ is increasing on $[0, \frac{3}{4}]$ (derivative $3-4u > 0$ there), so for $u \leq \frac{1}{4}$: $u(3-2u) \leq \frac{1}{4}(3 - \frac{1}{2}) = \frac{5}{8}$. Multiply by $U = un$: $g(U) = U \cdot u(3-2u) \leq \frac{5}{8}U$. $\square$

4 An elementary Chernoff bound

This is the only concentration tool used in the whole proof, and the only place beyond elementary combinatorics that real exponentials appear.

Lemma 4.1 (MGF of a sum of independent Bernoulli-type trials). *Let $Y_1, \dots, Y_n$ be independent $\{0, 1\}$ -valued random variables on a finite uniform probability space, with $\mu = \sum_v \mathbb{E}[Y_v]$. Then for every real $t \geq 0$,*

$$\mathbb{E}[\exp(tX)] \leq \exp(\mu(e^t - 1)), \quad X := \sum_v Y_v.$$

Proof. Write $p_v = \mathbb{E}[Y_v] \in [0, 1]$. Since $Y_v \in \{0, 1\}$, $\mathbb{E}[\exp(tY_v)] = (1 - p_v) \cdot 1 + p_v \cdot e^t = 1 + p_v(e^t - 1)$. By independence, $\mathbb{E}[\exp(tX)] = \prod_v \mathbb{E}[\exp(tY_v)] = \prod_v (1 + p_v(e^t - 1))$. By the elementary inequality $1 + y \leq e^y$ (valid for every real y), each factor is $\leq \exp(p_v(e^t - 1))$, so the product is $\leq \exp(\sum_v p_v(e^t - 1)) = \exp(\mu(e^t - 1))$. $\square$

Lemma 4.2 (Chernoff tail bound). *In the setting of Lemma 4.1, for every $t \geq 0$ and every $k \in \mathbb{R}$:*

$$\mathbb{P}[X \geq k] \leq \exp(\mu(e^t - 1) - tk).$$

Proof. Markov's inequality applied to the nonnegative random variable $\exp(tX)$ (using $t \geq 0$, so $X \geq k \iff \exp(tX) \geq \exp(tk)$): $\mathbb{P}[X \geq k] = \mathbb{P}[\exp(tX) \geq \exp(tk)] \leq \mathbb{E}[\exp(tX)] / \exp(tk) \leq \exp(\mu(e^t - 1) - tk)$ by Lemma 4.1. $\square$

Corollary 4.3 (Closed-form tail bounds). *In the setting of Lemma 4.1, for every $k \geq \mu > 0$, choosing $t = \log(k/\mu) \geq 0$ in Lemma 4.2 gives*

$$\mathbb{P}[X \geq k] \leq \exp\left(k - \mu - k \log \frac{k}{\mu}\right).$$

Symmetrically (applying the above to the independent family $1 - Y_v$, with sum $n - X$ and mean $n - \mu$), for every $0 < k \leq \mu$:

$$\mathbb{P}[X \leq k] \leq \exp\left(k - \mu + k \log \frac{\mu}{k}\right).$$

Proof. Substitute $t = \log(k/\mu)$ into Lemma 4.2: $\mu(e^t - 1) - tk = \mu(k/\mu - 1) - k \log(k/\mu) = k - \mu - k \log(k/\mu)$. For the second bound, apply the first to $X' := n - X = \sum_v (1 - Y_v)$ (mean $n - \mu$) and target $k' := n - k \geq n - \mu$: $\mathbb{P}[X \leq k] = \mathbb{P}[X' \geq k'] \leq \exp(k' - (n - \mu) - k' \log \frac{k'}{n - \mu})$; substituting $k' = n - k$ and simplifying (a routine but slightly tedious algebraic rewrite, since n cancels) yields $\exp(k - \mu + k \log(\mu/k))$. $\square$

Remark 4.4. *Both bounds are mean-scaled: the exponent depends on μ directly, not on the crude worst-case “variance proxy” $n/4$ that a generic Hoeffding bound for $[0, 1]$ -bounded variables would use. This matters: Corollary 4.3, applied with k a fixed small deviation from a small μ (as happens late in the saturation phase, once few dissenting agents remain), still gives a strong bound, whereas a range-based Hoeffding bound would be vacuous there — its implicit variance proxy $n/4$ never shrinks even as the true number of at-risk agents does. This is exactly the ingredient that lets the argument reach exact consensus ($U_T = 0$) rather than merely a $O(\sqrt{n \log n})$ -close approximation.*

5 Growth phase

Set $\beta_j = \frac{1}{10} \cdot (1.1)^j$ for $j = 0, \dots, 10$; note $\beta_0 = 0.1$ (matching the hypothesis $X_0 \geq 0.6n$) and $\beta_{10} = 0.1 \times 1.1^{10} = 0.259 \dots > 0.25$.

Lemma 5.1 (Phase 1). *There is an explicit $c_1 > 0$ such that: for every n , if $X_t \geq n(\frac{1}{2} + \beta_j)$ for some $j < 10$, then*

$$\mathbb{P}\left[X_{t+1} < n\left(\frac{1}{2} + \beta_{j+1}\right)\right] \leq \exp(-c_1 n).$$

Consequently, by a union bound over 10 rounds, starting from $X_0 \geq 0.6n$,

$$\mathbb{P}[X_{10} < 0.75n] \leq 10 \exp(-c_1 n).$$

Proof. By the reference-embedding trick (Lemma 1.4), it suffices to prove the bound for the reference set of size exactly $M = \lceil n(\frac{1}{2} + \beta_j) \rceil$ (worst case; the actual $X_t \geq M$ dominates it). By Lemma 3.1 and monotonicity of p (which follows since $p'(x) = 6x(1-x) \geq 0$), the mean of X_{t+1} from this reference set is $\mu := np(M/n) \geq np(\frac{1}{2} + \beta_j) \geq n(\frac{1}{2} + \frac{5}{4}\beta_j)$ by Corollary 3.3 (using $\beta_j \leq \beta_9 < \frac{1}{4}$). Set $k := n(\frac{1}{2} + 1.1\beta_j) < \mu$ (guaranteed factor 1.1, strictly below the true drift factor 1.25). Apply the lower-tail bound of Corollary 4.3 with these k, μ (using $\mu \geq n \cdot 0.5 + n \cdot 1.25 \times 0.1 = 0.625n$, in particular $\mu > 0$): a direct computation of $k - \mu + k \log(\mu/k)$ — using $k/\mu \leq \frac{0.5+0.11}{0.5+0.125} = 0.976$ at the (numerically worst, smallest- β_j) case $j = 0$ — gives an explicit negative constant times n ; taking c_1 to be (the absolute value of) this constant proves the single-round bound, uniformly over $j < 10$ since the bound only improves for larger β_j (the ratio k/μ moves further from 1). The union bound over the (at most) 10 rounds gives the stated total. $\square$

6 Saturation phase

Write $U_t = n - X_t$. By Lemma 5.1, $U_{10} \leq 0.25n$ except with probability $\leq 10 \exp(-c_1 n)$; we now show that, from this point, U_t reaches exactly 0 within $\lceil 6 \log n \rceil + 2$ further rounds, except with probability $O(1/n)$.

Stage 2a: geometric descent (via Corollary 3.4)

Lemma 6.1 (Stage 2a). *There is an explicit constant $c_2 > 0$ such that: whenever $462 \log n \leq U_t \leq 0.25n$,*

$$\mathbb{P}[U_{t+1} > 0.7U_t] \leq n^{-2}.$$

Consequently, starting from $U_{10} \leq 0.25n$, after $T_{2a} := \lceil 6 \log n \rceil$ further rounds, $U_{10+T_{2a}} \leq 462 \log n$ except with probability at most $T_{2a} \cdot n^{-2} \leq (6 \log n + 1)/n^2$.

Proof. By the reference-embedding trick applied to the dissenting set (the symmetric counterpart of Lemma 1.4, using (3)), it suffices to bound $\mathbb{P}[U' > 0.7M]$ where U' is the next-round dissent count starting from a reference set of exactly M dissenters, $U_t \leq M$. By Corollary 3.4, $\mu := \mathbb{E}[U'] = g(M) \leq \frac{5}{8}M$. Apply the upper-tail bound of Corollary 4.3 with $k = 0.7M \geq \frac{5}{8}M \geq \mu$: writing $\rho = k/\mu \geq 0.7/0.625 = 1.12$ (using the conservative substitution $\mu = \frac{5}{8}M$, which only decreases $\mu(\rho - 1 - \rho \log \rho)$'s magnitude compared to the true, possibly smaller, μ — a short calculus check that $h(\rho) := \rho - 1 - \rho \log \rho$ is decreasing for $\rho \geq 1$, so this is the least favorable case), the exponent evaluates to $\mu \cdot h(1.12) = \frac{5}{8}M \times (-0.00693 \dots)$; taking absolute value and clearing $M \geq 462 \log n$ gives exponent $\leq -2 \log n$, i.e. the bound n^{-2} , with c_2 chosen as $\frac{5}{8} \times 0.00693 \dots / 462$ made exact in the Lean development (the precise value of c_2 is immaterial; only the resulting inequality matters).

Iterate: while the current bound $M_j = 0.25n \cdot (0.7)^j$ remains $\geq 462 \log n$, each round contracts the reference bound by a factor 0.7 with failure probability $\leq n^{-2}$; a union bound over $T_{2a} = \lceil 6 \log n \rceil$ rounds gives total failure $\leq T_{2a} n^{-2}$, and $(0.7)^{T_{2a}} \cdot 0.25n \leq 462 \log n$ holds comfortably for $T_{2a} = \lceil 6 \log n \rceil$ (indeed $(0.7)^{6 \log n} = n^{6 \log 0.7} = n^{-2.14 \dots}$, so $(0.7)^{T_{2a}} \cdot 0.25n = O(n^{-1.14})$, far below $462 \log n$ for large n), which is what drives $U_{10+T_{2a}}$ below the $462 \log n$ floor. $\square$

Stage 2b: one round to a fixed constant

Lemma 6.2 (Stage 2b). *There is N_1 such that for $n \geq N_1$: whenever $U_t \leq 462 \log n$,*

$$\mathbb{P}[U_{t+1} > 10] \leq n^{-10}.$$

Proof. As before, reduce to the reference set of size $M = 462 \log n$ (worst case), with $\mu = g(M) \leq 3M^2/n = 3 \times 462^2 (\log n)^2/n$, a quantity that $\rightarrow 0$ as $n \rightarrow \infty$ (since $(\log n)^2 = o(n)$); fix N_1 large enough that $\mu \leq 1$ for $n \geq N_1$ (an explicit, if generous, threshold). Apply the upper-tail bound of Corollary 4.3 with $k = 10 \geq \mu$: the exponent $k - \mu - k \log(k/\mu) = 10 - \mu - 10 \log(10/\mu)$ is a continuous, strictly decreasing function of μ on $(0, 10]$ tending to $-\infty$ as $\mu \rightarrow 0^+$; in particular it is $\leq -10 \log n$ once μ is small enough, which holds for n larger than a second explicit threshold (again folded into N_1). This gives the stated n^{-10} bound (any polynomially-small target would do; n^{-10} is simply a convenient, generously small choice that leaves slack for all other error terms in the union bound of Theorem 2.1). $\square$

Stage 2c: the last round

Lemma 6.3 (Stage 2c). *Whenever $U_t \leq 10$: $\mathbb{P}[U_{t+1} \geq 1] \leq 300/n$.*

Proof. By the reference-embedding trick, reduce to $M = 10$: $\mu = g(10) = \frac{100}{n} (3 - \frac{20}{n}) \leq \frac{300}{n}$ (as $3 - 20/n < 3$). Since U_{t+1} is a nonnegative integer, ordinary Markov gives $\mathbb{P}[U_{t+1} \geq 1] \leq \mathbb{E}[U_{t+1}] = \mu \leq 300/n$. $\square$

7 Proof of Theorem 2.1

With $T = 10 + \lceil 6 \log n \rceil + 2$, write the failure event $\{I_T \neq V\} = \{U_T \geq 1\}$ as a union, via a union bound chained across the phases above (conditioning successively on each phase’s “good” event, exactly as in the push protocol’s `prNotAllInformed1e`):

$$\mathbb{P}[U_T \geq 1] \leq \underbrace{10e^{-c_1 n}}_{\text{Lemma 5.1}} + \underbrace{\frac{6 \log n + 1}{n^2}}_{\text{Lemma 6.1}} + \underbrace{n^{-10}}_{\text{Lemma 6.2}} + \underbrace{\frac{300}{n}}_{\text{Lemma 6.3}}.$$

For $n \geq N_0 := \max(N_1, \text{small explicit constants making the first three terms } \leq 1/n \text{ each})$, the right side is $\leq C/n$ for the explicit constant $C = 303$ (three “1’s” of slack from the first three terms, plus 300 from the last). $\square$

Remark 7.1 (On tightness). *As with the push protocol’s write-up, the constants here (the 60%/75% thresholds, the 10 growth rounds, the 462 and 10 in the saturation stages, the final $C = 303$) are chosen to make every inequality easy to verify by direct computation, not to be sharp. The clean statement to formalize is: for any fixed $\delta_0 \in (0, \frac{1}{20}]$, growth threshold $\beta^* \in (0, \frac{1}{2\sqrt{2}})$, contraction margin $\rho > 1$, and stage-2b target $k_0 \geq 1$, the same three-lemma structure goes through, and the specific numbers above are simply one admissible choice.*

8 The Lean formalization

The argument above is fully formalized (no `sorry`) in the accompanying repository (library `ThreeMajority`, namespace `ThreeMajority`). Design notes, by file:

- **Prob.lean**: the same finite-uniform `avg` and `expList` machinery as `rumor_spread/RumorSpread/Prob.1` (copied, since the two projects are independent Lean packages), plus one new general lemma `avg_prod_pi`: for a Pi-type $\iota \rightarrow \gamma$ of Fintypes, the average of a product $\prod_i f_i(x_i)$ over the product space equals the product of the individual averages $\prod_i \text{avg}(f_i)$ — the discrete, measure-theory-free form of independence of coordinate projections on a product space, proved by a direct sum/product manipulation (`Finset.prod_univ_sum`-style), with no appeal to Mathlib’s `MeasureTheory/PMF/ProbabilityTheory.iIndepFun` libraries.

- **Model.lean**: agents $\text{Fin } n$; a round configuration $\text{Tgt3 } n := \text{Fin } n \rightarrow \text{Fin } n \times \text{Fin } n \times \text{Fin } n$; **step** counts, per agent, how many of the three samples lie in the current opinion-1 set and compares to 2; **run** consumes a list of rounds. The monotone coupling of Lemma 1.4 is **step_mono**.
- **Chernoff.lean**: Lemma 4.1 through Corollary 4.3, entirely in the **avg** framework: the per-agent MGF identity uses only **Real.add_one_le_exp** (Mathlib’s $1 + x \leq e^x$) and **avg_prod_pi**; the closed-form tail bounds instantiate the free parameter **t** with an explicit **Real.log** term at each call site (mirroring how **RumorSpread/Main.lean** instantiates **L**, **T1**, **T2** with explicit **Real.log**-based numerals, rather than proving a fully generic “optimal t ” lemma).
- **OneRound.lean**: Lemma 3.1 and Lemma 3.2’s identities, as exact polynomial computations over the reals (**ring/nlinarith**), analogous in spirit to **RumorSpread/OneRound.lean**’s **avg_not_contacted** but simpler: no subtype/counting gymnastics are needed since samples may coincide or equal the sampling agent, so **Tgt3 n** is a plain iterated product rather than a dependent subtype as in **RumorSpread.Tgt**.
- **Growth.lean** / **Saturation.lean**: Lemma 5.1 and Lemmas 6.1–6.3 respectively, each a direct instantiation of Corollary 4.3 at explicit numeric parameters, glued across rounds by the same **expList**-conditioning idiom as **RumorSpread/Main.lean**’s **prNotAllInformed.le**.
- **Main.lean**: the numeric assembly (Section 3’s proof of Theorem 2.1) and the main theorem **majority3_consensus_whp**.
- **Design constraint preserved from rumor_spread**: no measure theory, no **PMF/ENNReal**, no Mathlib **ProbabilityTheory** kernels — everything is finite sums, plus **Real.exp/Real.log** where the Chernoff bound genuinely needs them (unlike the push protocol, where **Real.exp** appears only in a numeric side lemma, here it is structurally central, since a non-monotone process forces genuine concentration rather than a counting argument).

References

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- [2] L. Becchetti, A. Clementi, E. Natale, F. Pasquale, R. Silvestri. *Plurality consensus in the gossip model*. SODA 2015.